A Phase 1b, Open-Label, Multiple Ascending Dose Study to Investigate the Safety and Tolerability of OSA-1-100 Intravitreal Implant in Subjects with Dry Age-Related Macular Degeneration (AMD)

Protocol Number: OCS-1-002

Protocol Version:	1.0
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Sponsor:	Investigator sponsored study by Asociación para Evitar la Ceguera (APEC) Departamento de Investigación
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Principal Investigator: (Study Director)	Dr. Guillermo Salcedo Villanueva
Funding Provided By:	Osanni Bio, Inc.

SIGNATURE OF AGREEMENT FOR PROTOCOL OCS-1-002

I have read this protocol and agree to conduct the study as outlined herein in accordance with current Good Clinical Practice (GCP) and ICH Guidelines and in compliance with the obligations and requirements of clinical investigators set forth by local government agencies.

Dr. Guillermo Salcedo Villanueva
Principal Investigator Name

______________________________ ______________________________
Principal Investigator Signature Date

1. PROTOCOL SUMMARY

TITLE	A Phase 1b, Open-Label, Multiple Ascending Dose Study to Investigate the Safety and Tolerability of OSA-1-100 Intravitreal Implant in Subjects with Dry Age-Related Macular Degeneration (AMD)
PROTOCOL VERSION	0.4
SPONSOR	Asociación para Evitar la Ceguera (APEC) Departamento de Investigación, funding provided by Osanni Bio, Inc.
TEST ARTICLE	The investigational product, OSA-1-100, is a sterile, intravitreal (IVT) implant composed of a biodegradable, solid poly(lactic-co-glycolic acid) (PLGA) polymer sustained-release drug delivery matrix. Subjects will receive one of the following two planned dose levels of study drug: • Low Dose Level: 100 µg OSA-1-100 • High Dose Level: 200 µg OSA-1-100 Subjects will receive their assigned study drug implant in their study eye at Baseline, with a new implant being administered every 12 weeks (at Weeks 12, 24 and 36) during the treatment period. Subjects will receive a total of four study drug implants over the course of the study.
CONTROL ARTICLE	Not applicable
OBJECTIVE	The primary objective of this study is to evaluate the safety and tolerability of multiple administrations of OSA-1-100 IVT implants in subjects with dry age-related macular degeneration (AMD). Safety will be determined by assessing the incidence and severity of treatment emergent adverse events (TEAEs) related to the study drug implant as well as evaluating any changes in abbreviated physical exam findings, vital signs, visual acuity, IOP, slit lamp biomicroscopy findings and dilated ophthalmoscopy findings.
CLINICAL HYPOTHESIS	Treatment with OSA-1-100 IVT implants does not result in any additional serious adverse events or significantly greater number of adverse events than expected by the IVT injection procedure alone.
STUDY DESIGN	This is a Phase 1b, open-label, multiple ascending dose study examining the safety and tolerability of OSA-1-100 IVT implants. The study will consist of two dosing groups: a Sentinel ascending-dose group and a maximum tolerated dose (MTD) group. The Sentinel group will enroll up to 5 subjects into each of two planned sequential ascending dose levels of OSA-1-100. Data from the Sentinel group will be evaluated by the Data Safety Committee (DSC) to determine if it is safe to escalate to the next dose level and to determine the initial MTD of OSA-1-100.

	Once the initial MTD has been established, up to 20 additional subjects with geographic atrophy (GA) secondary to AMD will be enrolled into the MTD group to confirm the MTD of OSA-1-100. The DSC will review safety data throughout the study and manage dose escalation and enrollment. The Screening visit will be performed within 30 days prior to the Baseline (Day 1) visit. Data to be collected during Screening will include vital signs, abbreviated physical examination, best-corrected visual acuity (BCVA) in normal luminance (NL-BCVA) using the Early Treatment Diabetic Retinopathy Study (ETDRS) scale, slit-lamp biomicroscopy examination, intraocular pressure (IOP), dilated ophthalmoscopy examination, spectral domain optical coherence tomography (SD-OCT), fundus autofluorescence (FAF), color fundus photography (CFP) and fluorescein angiography (FA). After the Screening examination, relevant data will be reviewed to confirm study eligibility. Eligible subjects will return for the Baseline visit, at which time qualified subjects will be enrolled into the study. **Sentinel group:** Five (5) subjects will be enrolled into the Low Dose Level and receive the Low Dose OSA-1-100 IVT implant in the study eye at Baseline, with a new implant being administered every 12 weeks during the treatment period. Once all enrolled Low Dose Level subjects have received their first study drug implant and have completed the Week 4 visit, the DSC will review available safety data through at least Week 4. If the Low Dose Level is determined by the DSC to be safe and tolerable, enrollment of 5 additional subjects into the High Dose Level will be allowed. Once dose-escalation has been completed or has been terminated due to the occurrence of dose-limiting toxicities (DLTs), the DSC will evaluate all available safety data through Week 4 of the highest dose level administered. The DSC will then determine the initial MTD of OSA-1-100. **MTD group:** Up to 20 additional subjects with GA secondary to AMD will be enrolled into the MTD group and receive the MTD IVT implant in the study eye at Baseline, with a new study drug implant being administered every 12 weeks during the treatment period. **Both groups:** Safety and tolerability will be assessed throughout the Treatment and Follow-up Periods, while exploratory efficacy outcomes will be measured at selected time points as specified in the Schedule of Events and Exams. After completion of the 48-week treatment period, subjects

	will continue to be monitored for safety over the 12-week follow-up period. The final study follow-up visit will occur at Week 60.
NUMBER OF SUBJECTS	Up to 30
RANDOMIZATION	None
SUBJECT POPULATION	Men and women at least 55 years of age with a diagnosis of dry AMD
STUDY DURATION	The overall duration of the study will be up to 64 weeks (a Screening Period of up to 30 days, a 48-week Treatment Period and a 12-week Safety Follow-Up Period)
SUBJECT SELECTION CRITERIA	**INCLUSION CRITERIA** **Ocular Inclusions** 1. At least 5 intermediate to large (≥125 μm) drusen in each eye as determined by the investigator and confirmed by the Reading Center, with the exception of the low dose in the Sentinel group which does not require confirmation by the Reading Center. 2. MTD group only: Clinical diagnosis of GA secondary to AMD in both eyes as determined by the investigator and confirmed by the Reading Center. (There is no GA diagnosis requirement for the Sentinel group.) 3. MTD group only: GA area ≥1.0 and ≤7.5 mm^2 (0.4 and 3 disk areas [DA], respectively) in BOTH eyes as determined by the Investigator's assessment of Screening fundus autofluorescence (FAF) and confirmed by the Reading Center. The entire GA area must be located within the FAF imaging field. (There are no GA area requirement for the Sentinel group.) 4. Normal luminance BCVA (NL-BCVA) assessed by Early Treatment Diabetic Retinopathy Study (ETDRS) charts in the study eye at Screening and Baseline as follows: a. Sentinel group: ≤50 letters (Snellen equivalent of 20/100 or worse). b. MTD group: No upper or lower limit for NL-BCVA. c. When both eyes are eligible for the study, only one eye will be selected as the study eye. In that case, the eye with the worse NL-BCVA will be chosen as the study eye. If NL-BCVA is equal in both eyes, the study eye will be selected as follows: odd birth month – OD and even birth month – OS.

5. Sufficiently clear ocular media, adequate pupillary dilation, fixation to permit quality fundus imaging, and able to cooperate sufficiently for adequate ophthalmic visual function testing and anatomical assessment in each eye.
6. Have available OCT and/or FAF imaging records within a period of 1-6 months prior to Screening to allow for assessment of recent structural disease progression.

General Inclusions

7. Male or female ≥55 years of age.
8. Able to provide informed consent and willing to comply with all study visits and examinations.
9. Women of childbearing potential who are not pregnant or nursing, and who have a negative urine pregnancy test at Screening.

EXCLUSION CRITERIA

Ocular Exclusions – Study Eye

1. Ocular incisional surgery (including cataract surgery) in the study eye within 90 days prior to Screening, or anticipated need for such surgery during the study.
2. Previous intravitreal drug delivery (e.g., intravitreal corticosteroid injection, anti-angiogenic drugs or device implantation) in the study eye within 90 days prior to Screening.
3. History or presence of choroidal neovascularization (CNV) or exudative AMD in the study eye.
4. History or presence of retinal detachment in the study eye.
5. History or presence of diabetic retinopathy or diabetic macular edema (DME) in the study eye. (A history of diabetes mellitus without retinopathy or DME is allowed.)
6. Presence of retinal vein or artery occlusion impacting the macular area in the study eye.
7. MTD group only: History or presence of macular hole in the study eye.

Ocular Exclusions – Non-study (Fellow) Eye

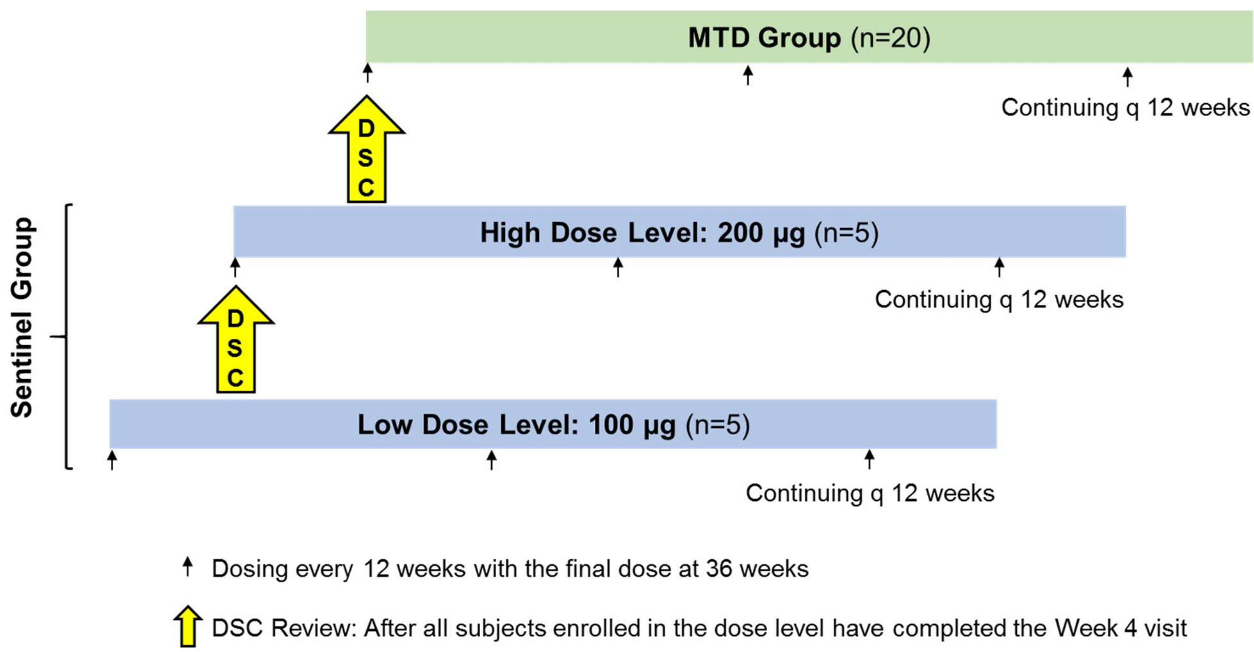
8. MTD group only: History or presence of choroidal neovascularization (CNV) or exudative AMD in the fellow eye.
9. MTD group only: History or presence of diabetic retinopathy or diabetic macular edema (DME) in the fellow eye. (A history of diabetes mellitus without retinopathy or DME is allowed.)
10. MTD group only: Presence of retinal vein or artery occlusion impacting the macular area in the fellow eye.

	Ocular Exclusions – Both Eyes 11. Active, infectious conjunctivitis, keratitis, scleritis or endophthalmitis. 12. Presence of significant keratopathy that would cause scattering of light or alter visual function, especially in mesopic (low luminance) conditions. 13. History or presence of advanced guttae indicative of Fuchs' endothelial dystrophy. 14. Chronic, uncontrolled glaucoma or ocular hypertension defined as IOP >25 mmHg with or without concomitant treatment with IOP-lowering medications. (Treatment with IOP-lowering medications is allowed during the study.) 15. Presence of visually significant cataract OR pseudophakia with evidence of posterior capsular opacity. 16. Aphakia. 17. Active uveitis and/or vitritis (grade trace or above). 18. History or presence of idiopathic or autoimmune-associated uveitis. 19. EPresence of vitreous hemorrhage. 20. Presence of visually or anatomically significant epiretinal membrane/macular pucker. 21. Prior treatment with Visudyne® (verteporfin), external-beam radiation therapy (for intraocular conditions), or transpupillary thermotherapy. 22. History of prophylactic subthreshold laser treatment for retinal disease. 23. History of vitrectomy surgery, submacular surgery, or any vitreoretinal surgery within 12 months prior to Screening, or anticipated need for such surgery during the study. 24. Treatment with pegcetacoplan or avacincaptad pegol within 6 months prior to Screening or during the study. 25. Inability to obtain color fundus photographs, FAF images and/or fluorescein angiography images of sufficient quality to be analyzed and interpreted. **General Exclusions** 26. Participation in another investigational drug or device clinical trial within 90 days prior to Screening or during the study. 27. History of allergy to fluorescein that is not amenable to treatment. 28. History of allergic reaction to OSA-1-100 or any of its components. 29. Inability to comply with study or follow-up procedures.

OUTCOME MEASURES	**Primary Safety Parameters** Safety will be determined by assessing the incidence and severity of treatment emergent adverse events (TEAEs) related to the study drug implant and the occurrence of DLTs. **Additional Safety Parameters** Additional safety parameters to be evaluated include changes in abbreviated physical examination findings, vital signs, visual acuity, IOP, slit lamp biomicroscopy findings and any findings from the dilated ophthalmoscopy examination. **Exploratory Efficacy Parameters** Exploratory efficacy parameters include: • Change from baseline in drusen volume • Change from baseline in peak/maximum drusen height • Change from baseline in area of geographic atrophy • Change from baseline in ellipsoid zone (EZ) thickness above peak drusen height • Change from baseline in outer nuclear layer (ONL) thickness above peak drusen height • Change from baseline in rate of photoreceptor (EZ) loss and EZ attenuation area in study eye vs fellow eye • Change from baseline in *en face* choroidal hypertransmission defects in study eye vs. fellow eye • Change from baseline in NL-BCVA • Change from Baseline in low luminance BCVA (LL-BCVA) in MTD subjects • Change from Baseline in Contrast Sensitivity in MTD subjects
SAMPLE SIZE JUSTIFICATION	The sample size was selected to place the minimum number of subjects at risk while providing sufficient safety information to move forward with subsequent larger studies.
STATISTICAL ANALYSIS PLAN	All study parameters will be listed, and a selected list of parameters will be summarized descriptively for each dose level and overall. The sample size, mean, standard deviation, median, minimum and maximum will be presented for continuous analysis variables. Categorical variables will be presented as the count and percentage for each category. Statistical tests will be two-sided and evaluated at an α of 0.05, unless otherwise specified. No adjustment will be made for multiplicity. Exploratory efficacy measures will be analyzed using paired statistics, paired t-test or Wilcoxon signed-rank test, comparing the study eye

	result to baseline or comparing the change from baseline between the study eye and the fellow eye.
DATA SAFETY COMMITTEE	A Data Safety Committee (DSC) has been appointed for this study. The DSC is a group of independent physicians and scientists who are appointed to monitor the safety and scientific integrity of a human research intervention, and to make recommendations to the sponsor regarding escalation to the next dose level or stopping the study for safety concerns. At each clinic visit during the study, subjects will undergo ocular evaluations to detect any signs of potential toxicity. For each of the two planned ascending dose levels of OSA-1-100 in the Sentinel group, the DSC will review data once all enrolled subjects at that dose level have received their first study drug implant and have completed the Week 4 visit. The DSC will evaluate available safety data through at least Week 4 before allowing dosing to escalate to the next dose level. Once dose-escalation has been completed or has been terminated due to the occurrence of dose-limiting toxicities (DLTs), the DSC will evaluate all available safety data through Week 4 of the highest dose level administered. The DSC will then determine the MTD of OSA-1-100 for administration in the MTD group of up to 20 additional eyes. If at any point there is a safety concern that emerges regarding the continued administration of OSA-1-100, the DSC will recommend the Sponsor to halt further dosing.

2. STUDY SCHEMA



3. SCHEDULE OF VISITS AND EXAMS

Assessment	Screening Period		Treatment Period			Follow-up Period	Early Termination Visit
	Screening Day -30 to Day -1	Baseline Day 1	Day 2[a] 24 hrs Post Injection (± 4 hrs)	Week 1 Day 8 (± 2 days)	Week 4, 8, 12[bc], 16, 20, 24[bc], 28, 32, 36[bc], 40, 44, 48[c] (± 4 days)	Week 60 Safety Follow Up (± 7 days)	
Informed consent	X						
Demographics	X						
Medical/Ocular history	X						
Concomitant medications	X	X	X	X	X	X	X
Vital signs	X	X		X	X	X	X
Abbreviated physical exam	X[d]	X				X	X
Urine pregnancy test[e]	X					X	X
NL-BCVA	OU	OU	SE	OU	OU	OU	OU
LL-BCVA (MTD subjects only)		OU			OU[c]	OU	OU
Contrast sensitivity (MTD subjects only)		OU			OU[c]	OU	OU
Slit lamp biomicroscopy exam	OU	OU	SE	OU	OU	OU	OU
IOP	OU	OU	SE	OU	OU	OU	OU
Dilated ophthalmoscopy exam	OU	OU	SE	OU	OU	OU	OU
SD-OCT	OU	OU			OU	OU	OU

Assessment	Screening Period		Treatment Period			Follow-up Period	Early Termination Visit
	Screening Day -30 to Day -1	Baseline Day 1	Day 2[a] 24 hrs Post Injection (± 4 hrs)	Week 1 Day 8 (± 2 days)	Week 4, 8, 12[bc], 16, 20, 24[bc], 28, 32, 36[bc], 40, 44, 48[c] (± 4 days)	Week 60 Safety Follow Up (± 7 days)	
FAF	OU	OU			OU	OU	OU
CFP	OU	OU			OU	OU	OU
FA	OU					OU	OU
Eligibility	X	X					
Administration of study drug IVT implant		SE			SE[b]		
Post-injection safety assessment with IOP		SE			SE[b]		
Study drug implant evaluation		SE	SE	SE	SE	SE	SE
AE assessment		X	X	X	X	X	X

a The Day 2 (24 Hours Post Injection) visit may be conducted in the clinic or by phone, at the investigator's discretion. If a clinic visit is not performed, the subject will be contacted by phone to update concomitant medications and query for any potential AEs. If an AE is reported, the subject should return for an unscheduled visit per the judgement of the investigator.

b OSA-1-100 IVT implant administration and post-injection safety assessment performed at Baseline (Day 1) and at Weeks 12, 24 and 36.

c LL-BCVA and Contrast Sensitivity performed in MTD subjects only at Baseline (Day 1) and at Weeks 12, 24, 36, 48 and 60.

d Height and weight measured at the Screening visit only.

e Urine pregnancy test for women of childbearing potential (WOCBP) only; additional urine pregnancy testing may be performed anytime during the study.

Abbreviations: AE = adverse event; BCVA = best-corrected visual acuity; CFP = color fundus photography; FA = Fluorescein angiography; FAF = fundus autofluorescence; Hrs = hours; IOP= intraocular pressure; IVT = intravitreal; LL-BCVA = low luminance BCVA; NL-BCVA = normal luminance BCVA; OU = both eyes; SD-OCT = spectral-domain optical coherence tomography; SE = study eye.

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5. ABBREVIATION TABLE

Abbreviation	Definition
AE	Adverse event
ADR	Adverse drug reaction
AMD	Age related macular degeneration
APOE4	Apolipoprotein E4
BCVA	Best corrected visual acuity
CFP	Color fundus photography
CS	Clinically significant
DME	Diabetic macular edema
DLT	Dose limiting toxicity
DSC	Data Safety Committee
EC	Ethics Committee
eCRF	Electronic case report form
ETDRS	Early Treatment Diabetic Retinopathy Study
EZ	Ellipsoid zone
FA	Fluorescein angiography
FAF	Fundus autofluorescence
FDA	Food and Drug Administration
GA	Geographic atrophy
GCP	Good Clinical Practice
HDL	High-density lipoprotein
HEENT	Head, eyes, ears, nose and throat
HPMC	Hydroxypropyl methylcellulose
ICH	International Conference on Harmonisation
ID	Identification
LDL	Low-density lipoprotein
LL-BCVA	Low luminance best corrected visual acuity
MTD	Maximum Tolerated Dose
NCS	Not clinically significant
nm	Nanometer
NL-BCVA	Normal luminance best corrected visual acuity
NOAEL	No observed adverse effect level

Abbreviation	**Definition**
ONL	Outer nuclear layer
PLGA	Poly(lactic-co-glycolic acid)
RPE	Retinal pigment epithelium
SAE	Serious adverse event
SD-OCT	Spectral-domain optical coherence tomography
SUSAR	Serious and unexpected suspected adverse reaction
VEGF	Vascular endothelial growth factor
WOCBP	Women of childbearing potential

6. INTRODUCTION AND RATIONALE

6.1. AGE-RELATED MACULAR DEGENERATION (AMD)

Age-related macular degeneration (AMD) is defined by gradual degenerative changes in the macula, a small region in the center of the retina that is responsible for central vision. AMD is a disease that typically affects the elderly and is the major contributor to vision loss in those over 50 in developed countries (VanNewkirk et al., 2000). It is estimated that around 20 million people in the United States and 200 million people worldwide are currently affected by AMD (Rein et al., 2022), as the life expectancy in both developed and developing countries is rising (Ortman et al., 2014).

Early signs of AMD include drusen and abnormalities of the retinal pigment epithelium (RPE). The non-neovascular ("atrophic, dry") or neovascular ("wet or exudative") forms of AMD often progress as the illness advances with age to the advanced stage (Mitchell et al., 2018). The loss of photoreceptors, RPE cells, and accompanying capillaries (choriocapillaris) in the macula causes a noticeable thinning and/or atrophy of retinal tissue in the dry, atrophic form. Geographic atrophy (GA) is the name used for this overall pattern in the advanced stage of dry AMD (Holz et al., 2014). Additionally, GA that develops or progresses over time is a frequent cause of vision loss in patients with the neovascular or wet form of advanced AMD that are receiving anti-VEGF therapy, suggesting that GA is the common pathway to vision loss in most patients with AMD (Bhisitkul et al., 2015; Grunwald et al., 2017; Abdelfatah et al., 2016).

Geographic atrophy can result in bilateral, irreversible, and severe loss of functional vision (Mitchell et al., 2018; Sivaprasad et al., 2019; Boyer et al., 2017, Lindblad A et al., 2009), and is associated with a significant impact on quality of life and independence. In 2023, two anti-complement therapies were FDA-approved for the treatment of GA but their effect was modest in decreasing lesion progression at 24 months, and without any evidence of functional benefit. For the constantly growing aged population, the development of therapy options for geographic atrophy secondary to AMD is an area of urgent unmet medical need and a significant public health concern.

6.2. SCIENTIFIC RATIONALE FOR OSA-1-001

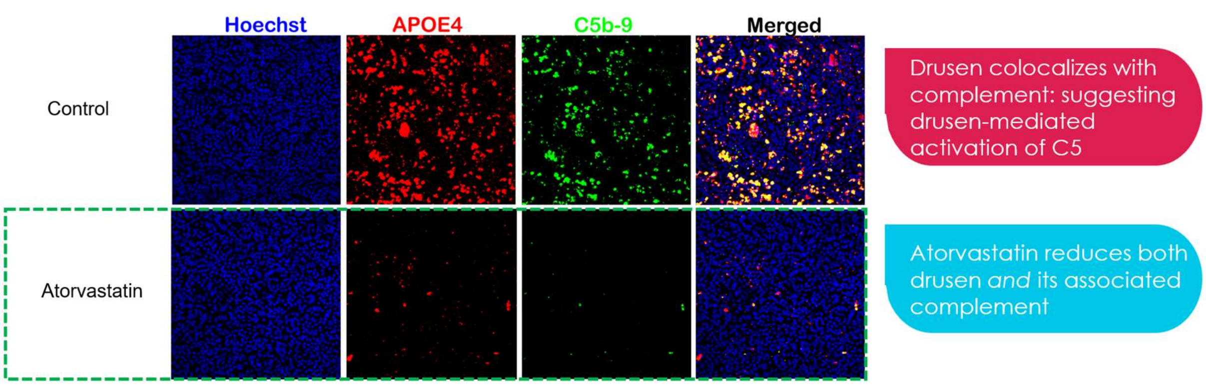
Drusen are the hallmark of AMD, and are known to be pathogenic in the AMD disease process. Large drusen are the largest single predictor of an adult progressing to late-stage AMD (either GA or neovascular AMD), conferring a 25% risk of late stage disease within 2 years (Folgar et al., 2016). In intermediate AMD, drusen height is very highly correlated with photoreceptor loss (Sadigh et al., 2013). For late-stage disease, the presence of large drusen confer a 30% risk of progression to neovascular AMD within 5 years (Bressler et al., 1990) and GA lesions form in the areas of prior large drusen (Gass, 1973).

Drusen have multiple pathogenic mechanisms that contribute to the progression of AMD. The primary mechanism is understood to be the creation of a diffusion barrier for oxygen and nutrient flow from the choroid to the RPE and retina (Sadigh et al., 2013; Bressler et al., 1990; Gass, 1973). Other secondary mechanisms include induction of VEGF secretion from hypoxic tissue and mitochondrial dysfunction (Hadziahmetovic and Malek, 2021; Curcio, 2018; Linsenmeier and Zhang, 2017), and drusen serve as a primary substrate for activation of the complement system (Johnson et al., 2011). As such, reducing drusen in the retina promises to be therapeutic with the potential to reduce the loss of photoreceptors, reduce the risk of progression to GA and/or wet AMD and potentially restore lost vision by improving oxygenation and nutrient supply to remaining retina.

Drusen are deposits of extracellular material between the basal lamina of the RPE and Bruch's membrane (Curcio, 2018). The principal component of the extracellular material are low-density lipoprotein (LDL)-like lipoprotein protein particles harboring cholesterol (Curcio, 2018). These cholesterol lipoprotein particles are large in diameter (approximately 100 nM) (Curcio, 2018) and get stuck when Bruch's membrane becomes less permeable to large particles with age (Moore and Clover, 2001). A proposed therapeutic strategy for reducing the size of drusen deposits is to inhibit the production of cholesterol by the RPE, thereby reducing the rate of deposition of lipoproteins. Statins, which are inhibitors of the enzyme HMG-CoA reductase, are widely used as an oral medication to decrease cholesterol production by the liver (Stancu and Sima, 2001). High dose oral administration of atorvastatin (Bannerman, 2023) and simvastatin (Guymer et al., 2013) have been used for AMD and were shown to reduce drusen volume and the risk of AMD progression in clinical trials. While the data on oral statins is encouraging, only 3% of the statin concentration in blood plasma is present in the vitreous.

Osanni Bio has shown in a human embryonic stem-cell-derived RPE (hESC-RPE) cell model that doses of statin that cannot be achieved through oral administration may be more effective at reducing drusen. In an Ossani Bio internal study, hESC-RPE cells cultured on transwell membrane inserts were treated with atorvastatin or control prior to addition of 5% human serum (a stimulator of drusen formation). The membranes were collected, and efficacy was evaluated via immunostaining for apolipoprotein E4 (APOE4) and C5b-9, the indicators of lipoprotein particles in drusen and drusen-mediated complement activation, respectively (see Figure 1). Staining results demonstrated that atorvastatin treatment remarkably decreased drusen deposits in the hESC-RPE cells, with reductions of APOE4 and complement factor C5b expression when compared to the control.

Figure 1: *In Vitro* Study of Drusen Formation



Stains used: Hoechst (RPE nuclei stain), APOE4 (ApoE lipoprotein stain for drusen), C5b-9 (complement C5 stain) and Merged (all stains).

Further supporting the utility of local statin delivery for AMD, topical ocular administration of atorvastatin encapsulated in solid lipid nanoparticles resulted in a meaningful reduction of drusenoid deposits in mice (Barathi et al., 2023).

Osanni Bio has developed OSA-1-100 as a sustained-release drug delivery formulation of atorvastatin suitable for implant into the vitreous. OSA-1-100 is being investigated as a potential treatment for dry AMD by reducing the production of cholesterol in the retina, thereby reducing drusen size and preventing or delaying disease progression.

6.3. Prior Clinical Trials

To date, no human clinical trials have been conducted with intravitreally administered OSA-1-100. However, oral administration of atorvastatin, the active ingredient in OSA-1-100, has been investigated in numerous clinical trials, and is currently approved in many countries at doses of up to 80 mg/day.

In the United States, atorvastatin is approved for the following indications (Lipitor® prescribing information, Viatris Specialty LLC):

- To reduce the risk of:
    - Myocardial infarction (MI), stroke, revascularization procedures, and angina in adults with multiple risk factors for coronary heart disease (CHD) but without clinically evident of CHD.
    - MI and stroke in adults with type 2 diabetes mellitus with multiple risk factors for CHD but without clinically evident CHD.
    - Non-fatal MI, fatal and non-fatal stroke, revascularization procedures, hospitalization for congestive heart failure, and angina in adults with clinically evident CHD.
- As an adjunct to diet, to reduce low-density lipoprotein cholesterol (LDL-C) in:
    - Adults with primary hyperlipidemia.
    - Adults and pediatric patients aged 10 years and older with heterozygous familial hypercholesterolemia.
- As an adjunct to other LDL-C-lowering therapies to reduce LDL-C in adults and pediatric patients aged 10 years and older with homozygous familial hypercholesterolemia.
- As an adjunct to diet for the treatment of adults with primary dysbetaliproteinemia or hypertriglyceridemia

The most commonly reported adverse reactions in patients treated with oral atorvastatin in clinical trials include the following: nasopharyngitis, arthralgia, diarrhea, pain in extremity, urinary tract infection, dyspepsia, nausea, musculoskeletal pain, muscle spasms, myalgia, insomnia and pharyngolaryngeal pain. Other important adverse reactions in atorvastatin-treated patients include myopathy, rhabdomyolysis, hepatic dysfunction, and increases in HbA1c and fasting serum glucose levels.

6.4. Dose Selection and Margin of Safety

The dose selection for the initial clinical study is based on the results of an ongoing single/repeat dose toxicology study performed in Dutch Belted Rabbits (Osanni Bio Internal Study 23-1475). To date all animals have received a single dose of OSA-1-100, consisting of a rod implant injected into the vitreous of each eye containing 200 µg/eye of atorvastatin mixed with various PLGA polymers. Using an average volume of 1.5 mL for the rabbit vitreous, each animal received a calculated total dose of 133.3 µg/mL/eye of atorvastatin. The actual vitreous concentration over time is dependent on the release characteristics of the polymers used in the implants.

Available toxicology results through 90 days indicate that all doses are safe, with no adverse test article related ocular or systemic effects observed in the rabbits. Based on these results, the NOAEL for intravitreally administered atorvastatin is established at 133.3 µg/mL/eye.

Using an average volume of 4.8 mL for the vitreous of the human eye, an implant with 200 µg of atorvastatin would result in a calculated total dose of 41.7 µg/mL/eye, yielding a safety margin of 3.2 relative to the NOAEL in rabbits.

6.5. Benefit/Risk Assessment

The potential benefit of OSA-1-100 is unknown. If efficacious, it may decrease drusen and may reduce the rate of increase in geographic atrophy area. There are known risks associated with the intravitreal route of administration including pain, infection (endophthalmitis), decreased or loss of vision, retinal tear/detachment, retinal pigment epithelial tear, intraocular inflammation, transient increase in intraocular pressure, subconjunctival or vitreous hemorrhage, and corneal abrasion or irritation.

More detailed information about the known and expected benefits and risks and reasonably expected adverse events for OSA-1-100 may be found in the Investigator’s Brochure.

7. STUDY OBJECTIVE

The primary objective of this study is to evaluate the safety and tolerability of multiple administrations of OSA-1-100 IVT implants in subjects with dry age-related macular degeneration (AMD).

Safety will be determined by assessing the incidence and severity of treatment emergent adverse events (TEAEs) related to the study drug implant as well as evaluating any changes in abbreviated physical exam findings, vital signs, visual acuity, IOP, slit lamp biomicroscopy findings and dilated ophthalmoscopy findings.

8. STUDY PLAN

8.1. Study Design

This is a Phase 1b, open-label, multiple ascending dose study examining the safety and tolerability of OSA-1-100 IVT implants. The study will consist of two dosing groups: a Sentinel ascending-dose group and a maximum tolerated dose (MTD) group. The Sentinel group will enroll up to 5 subjects into each of two planned sequential ascending dose levels of OSA-1-100. Data from the Sentinel group will be evaluated by the Data Safety Committee (DSC) to determine if it is safe to escalate to the next dose level and to determine the initial MTD of OSA-1-100. Once the initial MTD has been established, up to 20 additional subjects with geographic atrophy (GA) secondary to AMD will be enrolled into the MTD group to confirm the MTD of OSA-1-100. The DSC will review safety data throughout the study and oversee dose escalation and enrollment (see Section 2: Study Schema).

The Screening visit will be performed within 30 days prior to the Baseline (Day 1) visit. Data to be collected during Screening will include vital signs, abbreviated physical examination, best-corrected visual acuity (BCVA) in normal luminance (NL-BCVA) using the Early Treatment Diabetic Retinopathy Study (ETDRS) scale, slit-lamp biomicroscopy examination, intraocular pressure (IOP), dilated ophthalmoscopy examination, spectral domain optical coherence tomography (SD-OCT), fundus autofluorescence (FAF), color fundus photography (CFP) and fluorescein angiography (FA). After the Screening examination, relevant data will be reviewed to confirm study eligibility. Eligible subjects will return for the Baseline visit, at which time qualified subjects will be enrolled into the study.

Sentinel group:

Five (5) subjects will be enrolled into the Low Dose Level and receive the Low Dose OSA-1-100 IVT implant in the study eye at Baseline, with a new implant being administered every 12 weeks during the treatment period. Once all enrolled Low Dose Level subjects have received their first study drug implant and have completed the Week 4 visit, the DSC will review available safety data through at least Week 4. If the Low Dose Level is determined by the DSC to be safe and tolerable, enrollment of 5 additional subjects into the High Dose Level will be allowed.

Once dose-escalation has been completed, or has been terminated due to the occurrence of dose-limiting toxicities (DLTs), the DSC will evaluate all available safety data through Week 4 of the highest dose level administered. The DSC will then determine the initial MTD of OSA-1-100.

MTD group:

Up to 20 additional subjects with GA secondary to AMD will be enrolled into the MTD group and receive the MTD IVT implant in the study eye at Baseline, with a new study drug implant being administered every 12 weeks during the treatment period.

Both groups:

Safety and tolerability will be assessed throughout the Treatment and Follow-up Periods, while exploratory efficacy outcomes will be measured at selected time points as specified in the Schedule of Events and Exams. After completion of the 48-week treatment period, subjects will continue to be monitored for safety over the 12-week follow-up period. The final study follow-up visit will occur at Week 60.

Study procedures, their timing, and additional details are found in Section 3: Schedule of Visits and Exams and in Appendix A: Exams and Procedures.

8.2. Safety Monitoring

8.2.1. Dose-Limiting Toxicities

The following ocular AEs will be considered as dose-limiting toxicities (DLTs) if they are considered by the investigator to be related to the study drug:

- NL-BCVA loss from baseline of ≥15 ETDRS letters, unless the loss of vision is due to natural progression of the disease or is related to the injection procedure
- Anterior chamber cell and/or flare grade of +2 or greater
- Formation or worsening of cataract that is related to the study drug and not the injection procedure
- Increase from the pre-injection IOP of ≥15 mmHg lasting 60 or more minutes post-injection that is related to the study drug and not the injection procedure
- A sustained IOP of ≥30 mmHg lasting for more than 7 days despite pharmacologic intervention that is related to the study drug and not the injection procedure
- Vitreous hemorrhage that is related to the study drug and not the injection procedure
- Vitritis
- Retinal or choroidal inflammation
- Endophthalmitis

8.2.2. Data Safety Committee

A Data Safety Committee (DSC) has been appointed for this study. The DSC is a group of independent physicians and scientists who are appointed to monitor the safety and scientific integrity of a human research intervention, and to make recommendations to the sponsor regarding escalation to the next dose level or stopping the study for safety concerns.

At each clinic visit during the study, subjects will undergo ocular evaluations to detect any signs of potential toxicity. For each of the two planned ascending dose levels of OSA-1-100 in the Sentinel group, the DSC will review data once all enrolled subjects at that dose level have received their first study drug implant and have completed the Week 4 visit. The DSC will evaluate available safety data through at least Week 4 before allowing dosing to escalate to the next dose level.

If any DLT events are experienced by ≥3 subjects receiving the same dose level of OSA-1-100, enrollment of new subjects into the study will be suspended. Resumption of study enrollment may occur only with the approval of the DSC following additional review of safety data to determine if any changes in the study are necessary. If DLTs are observed, a dose intermediate between the tolerated dose in the lower study drug level and the dose associated with the DLTs may be evaluated.

Once dose-escalation has been completed or has been terminated due to the occurrence of dose-limiting toxicities (DLTs), the DSC will evaluate all available safety data through Week 4 of the highest dose level administered. The MTD will be determined as the dose level below the one at which DLT events are experienced by ≥2 subjects. If the dose-escalation scheme is successfully completed without achieving DLT termination criteria, the MTD will be defined as the highest dose level in which at most 1 subject experienced a DLT.

The DSC will continue to monitor subject safety data on a regular basis as described in the DSC Charter. If at any point there is a safety concern that emerges regarding the continued administration of OSA-1-100, the DSC will recommend the Sponsor to halt further dosing.

9. STUDY POPULATION

This study will enroll a total of up to 30 men and women at least 55 years of age with a diagnosis of dry AMD. For the study, only one eye of each eligible subject will be designated as the study eye; however, ophthalmic testing will be performed on both eyes as specified in the Schedule of Visits and Exams.

The study will be conducted at a single center in Mexico.

9.1. Inclusion Criteria

Potential subjects must meet all of the following criteria to be eligible for inclusion in the study:

Ocular Inclusions

1. At least 5 intermediate to large (≥125 µm) drusen in each eye as determined by the investigator and confirmed by the Reading Center, with the exception of the low dose in the Sentinel group which does not require confirmation by the Reading Center.
2. MTD group only: Clinical diagnosis of GA secondary to AMD in both eyes as determined by the investigator and confirmed by the Reading Center. (There is no GA diagnosis requirement for the Sentinel group.)
3. MTD group only: GA area ≥1.0 and ≤7.5 mm^2 (0.4 and 3 disk areas [DA], respectively) in BOTH eyes as determined by the Investigator's assessment of Screening fundus autofluorescence (FAF) and confirmed by the Reading Center. The entire GA area must be located within the FAF imaging field. (There are no GA area requirement for the Sentinel group.)
4. Normal luminance BCVA (NL-BCVA) assessed by Early Treatment Diabetic Retinopathy Study (ETDRS) charts in the study eye at Screening and Baseline as follows:
    a. Sentinel group: ≤50 letters (Snellen equivalent of 20/100 or worse).
    b. MTD group: No upper or lower limit for NL-BCVA.
    c. When both eyes are eligible for the study, only one eye will be selected as the study eye. In that case, the eye with the worse NL-BCVA will be chosen as the study eye. If NL-BCVA is equal in both eyes, the study eye will be selected as follows: odd birth month – OD and even birth month – OS.

5. Sufficiently clear ocular media, adequate pupillary dilation, fixation to permit quality fundus imaging, and able to cooperate sufficiently for adequate ophthalmic visual function testing and anatomical assessment in each eye.
6. Have available OCT and/or FAF imaging records within a period of 1-6 months prior to Screening to allow for assessment of recent structural disease progression.

General Inclusions

7. Male or female ≥55 years of age.
8. Able to provide informed consent and willing to comply with all study visits and examinations.
9. Women of childbearing potential who are not pregnant or nursing, and who have a negative urine pregnancy test at Screening.

9.2. Exclusion Criteria

Potential subjects meeting any of the following criteria will be excluded from the study:

Ocular Exclusions – Study Eye

1. Ocular incisional surgery (including cataract surgery) in the study eye within 90 days prior to Screening, or anticipated need for such surgery during the study.
2. Previous intravitreal drug delivery (e.g., intravitreal corticosteroid injection, anti-angiogenic drugs or device implantation) in the study eye within 90 days prior to Screening.
3. History or presence of choroidal neovascularization (CNV) or exudative AMD in the study eye.
4. History or presence of retinal detachment in the study eye.
5. History or presence of diabetic retinopathy or diabetic macular edema (DME) in the study eye. (A history of diabetes mellitus without retinopathy or DME is allowed.)
6. Presence of retinal vein or artery occlusion impacting the macular area in the study eye.
7. MTD group only: History or presence of macular hole in the study eye.

Ocular Exclusions – Non-study (Fellow) Eye

8. MTD group only: History or presence of choroidal neovascularization (CNV) or exudative AMD in the fellow eye.
9. MTD group only: History or presence of diabetic retinopathy or diabetic macular edema (DME) in the fellow eye. (A history of diabetes mellitus without retinopathy or DME is allowed.)
10. MTD group only: Presence of retinal vein or artery occlusion impacting the macular area in the fellow eye.

Ocular Exclusions – Both Eyes

11. Active, infectious conjunctivitis, keratitis, scleritis or endophthalmitis.
12. Presence of significant keratopathy that would cause scattering of light or alter visual function, especially in mesopic (low luminance) conditions.
13. History or presence of advanced guttae indicative of Fuchs' endothelial dystrophy.
14. Chronic, uncontrolled glaucoma or ocular hypertension defined as IOP >25 mmHg with or without concomitant treatment with IOP-lowering medications. (Treatment with IOP-lowering medications is allowed during the study.)
15. Presence of visually significant cataract **OR** pseudophakia with evidence of posterior capsular opacity.
16. Aphakia.
17. Active uveitis and/or vitritis (grade trace or above).
18. History or presence of idiopathic or autoimmune-associated uveitis.
19. Presence of vitreous hemorrhage.
20. Presence of visually or anatomically significant epiretinal membrane/macular pucker.
21. Prior treatment with Visudyne® (verteporfin), external-beam radiation therapy (for intraocular conditions), or transpupillary thermotherapy.
22. History of prophylactic subthreshold laser treatment for retinal disease.
23. History of vitrectomy surgery, submacular surgery, or any vitreoretinal surgery within 12 months prior to Screening, or anticipated need for such surgery during the study.
24. Treatment with pegcetacoplan or avacincaptad pegol within 6 months prior to Screening or during the study.
25. Inability to obtain color fundus photographs, FAF images and/or fluorescein angiography images of sufficient quality to be analyzed and interpreted.

General Exclusions

26. Participation in another investigational drug or device clinical trial within 90 days prior to Screening or during the study.
27. History of allergy to fluorescein that is not amenable to treatment.
28. History of allergic reaction to OSA-1-100 or any of its components.
29. Inability to comply with study or follow-up procedures.

10. STUDY OUTCOME MEASUREMENTS

10.1. Primary Safety Parameters

Safety will be determined by assessing the incidence and severity of treatment emergent adverse events (TEAEs) related to the study drug implant and the occurrence of DLTs.

10.2. Additional Safety Parameters

Additional safety parameters to be evaluated include changes in abbreviated physical examination findings, vital signs, visual acuity, IOP, slit lamp biomicroscopy findings and any findings from the dilated ophthalmoscopy examination.

10.3. Exploratory Efficacy Parameters

Exploratory efficacy parameters include:

- Change from baseline in drusen volume
- Change from baseline in peak/maximum drusen height
- Change from baseline in area of geographic atrophy
- Change from baseline in ellipsoid zone (EZ) thickness above peak drusen height
- Change from baseline in outer nuclear layer (ONL) thickness above peak drusen height
- Change from baseline in rate of photoreceptor (EZ) loss and EZ attenuation area in the study eye vs fellow eye
- Change from baseline in *en face* choroidal hypertransmission defects in study eye vs. fellow eye
- Change from baseline in NL-BCVA
- Change from Baseline in low luminance BCVA (LL-BCVA) in MTD subjects
- Change from Baseline in Contrast Sensitivity in MTD subjects

11. STUDY TREATMENTS

11.1. OSA-1-100

The investigational product, OSA-1-100, is a sterile, intravitreal (IVT) implant containing 100 or 200 µg of the active agent, atorvastatin calcium, dispersed in a biodegradable, solid poly(lactic-co-glycolic acid) (PLGA) polymer sustained-release drug delivery matrix.

11.2. Study Drug Packaging, Labeling and Storage

Study drug (OSA-1-100) consists of an atorvastatin calcium implant assembled in a single-use injection device with a 25G needle and a hydroxypropyl methylcellulose (HPMC) polymer implant retention system. The OSA-1-100 implant in device is supplied in a sealed aluminum foil pouch (i.e., study drug kit). Study drug should be stored refrigerated at 2ºC to 8ºC.

The label on the study drug kit specifies the kit number, lot number and includes the statement "For Investigational Use Only" in Spanish.

Storage should be secure to restrict access of the study drug to only appropriate study staff.

Records of actual storage conditions (e.g., 24-hour or daily refrigerator temperature log) must be maintained. If there is a storage temperature excursion during the study (e.g., study drug refrigerator temperature is below 2ºC or above 8ºC, the affected study drug must be immediately quarantined until it can be confirmed with the drug manufacturer (Osanni Bio) that it is acceptable for continued use in the study. Temperature excursions should be documented along with the final decision regarding the study drug and any corrective action that was taken.

11.3. Supply of Study Drug

Study drug will be provided to the investigator from the drug manufacturer, Osanni Bio, Inc. Prior to the start of the study, the site will receive an initial supply of study drug kits. Upon receipt at the site, the study drug must be immediately stored refrigerated at 2ºC to 8ºC.

Resupply of study drug will occur periodically as needed throughout the study treatment period, depending on subject enrollment and study progress.

11.4. Study Drug Accountability

The investigator is required to document the receipt, dispensation, and return/destruction of all study drug (OSA-1-100) supplies provided by the drug manufacturer, Osanni Bio, in accordance with GCP, International Conference on Harmonisation (ICH) guidelines and institutional policy.

All used and unused study drug kits must be retained for drug accountability. After the Site Monitor has completed final accountability of study drug, and has provided written authorization for destruction, the used and/or unused OSA-2-100 kits will be disposed of/destroyed in accordance with institutional policies and procedures for the disposal of medical/biohazardous waste. Documented evidence of the destruction (whether destruction is on site or by a third-party biological waste disposal service) must be retained.

11.5. Subject ID Assignment, Randomization and Masking

Screening subjects that have provided written informed consent will be assigned a unique five-digit screening number consisting of a two-digit site number (e.g., 01) followed by a three-digit sequential subject number (e.g., 01-301, 01-302, 01-303, etc.).

If a potential study subject has provided written informed consent, met all eligibility criteria and agreed to participate in the study, the subject will be enrolled in the study and assigned to a study drug dose level. The subject’s screening number will be used as the subject ID number throughout the study. Patients in the Sentinel group who receive a dose level assignment but do not receive any administrations of study drug may be replaced in order to ensure that the minimum number of subjects is enrolled per dose level.

The study will be conducted in an open-label manner. Randomization and masking are not applicable.

11.6. Study Drug Dose Level Assignments

Eligible subjects will be assigned to one of two dosing groups: the Sentinel ascending-dose group or the maximum tolerated dose (MTD) group. Enrollment into the Sentinel group will be completed first, followed by enrollment into the MTD group.

Within the Sentinel group, subjects will receive one of the following two planned dose levels of study drug, which will be enrolled in ascending order by dose as described in the Study Schema:

- Low Dose Level: 100 µg OSA-1-100 (5 planned subjects)
- High Dose Level: 200 µg OSA-1-100 (5 planned subjects)

Subjects will receive their assigned study drug implant in their study eye at Baseline, with a new implant being administered every 12 weeks (at Weeks 12, 24 and 36) during the treatment period. Subjects will receive a total of four study drug implants over the course of the study.

The low dose level will be evaluated for clinical safety by the DSC before escalating to the high dose level. Once dose-escalation has been completed, or has been terminated due to the occurrence of DLTs, the DSC will evaluate all available safety data through Week 4 of the highest dose administered and determine the initial MTD of study drug to be used for the MTD group. Up to 20 additional subjects will be enrolled into the MTD group.

11.7. Study Drug Preparation

Prior to study drug administration, OSA-1-100 kits must be removed from the refrigerator for at least 60 minutes to allow the study drug to reach room temperature. Once an OSA-1-100 kit has been completely warmed to room temperature, it may NOT be re-refrigerated for later use.

Label the OSA-1-100 foil pouch with the Subject ID and the Date of Injection prior to administration.

11.8. Study Drug Administration

The study drug will be administered only to subjects who have signed and dated an Informed Consent Form and are approved for enrollment into this study.

Just prior to administration, the foil pouch containing the preloaded study drug syringe should be opened, and the syringe placed onto a sterile field. Study drug is to be administered by IVT injection under aseptic conditions in accordance with institutional requirements and best practices. Using aseptic technique, the OSA-1-100 implant will be injected from the preloaded syringe into the vitreous of the study eye. Adequate topical anesthesia and betadine at the injection site will be administered prior to the injection procedure.

All used study drug kits (i.e., opened foil pouches) must be retained for drug accountability purposes.

The used injection syringes should be disposed of/destroyed in accordance with institutional policies and procedures for the disposal of medical/biohazardous waste.

Following each IVT injection of study drug, subjects will be monitored for elevation of IOP, decreased optic nerve head perfusion, and for possible injection complications (e.g., vitreous hemorrhage, retinal tears, etc.). Additionally, subjects will be instructed to immediately report any symptoms suggestive of endophthalmitis, such as ocular pain, swelling, redness, haze and gradual loss of vision.

Subjects may be prescribed antibiotic eye drops and/or preservative-free artificial tears following administration of study drug per the investigator's standard practice for IVT injections. Antibiotic drops and artificial tears, if prescribed, must be documented in the subject's electronic case report form (eCRF).

12. STUDY PROCEDURES AND ASSESSMENTS

The overall duration of the study will be up to 64 weeks (a Screening Period of up to 30 days, a 48-week Treatment Period and a 12-week Safety Follow-Up Period). The schedule of study visits and exams can be found in Section 3, and detailed descriptions of exams and procedures performed during these visits can be found in Appendix A. A detailed overview of the assessments performed at each study visit is presented below.

Note: The protocol-specified procedures for a given study visit may be split across 2 or more days as long as all of the procedures for that visit are completed within the specified visit window.

12.1. SCREENING VISIT (DAY -30 TO -1)

At the Screening visit, potential subjects will be evaluated for eligibility. Prior to performing any study specific procedures, potential subjects must provide written informed consent using the current Ethics Committee (EC)-approved informed consent form. Subjects officially enter the Screening Period following provision of informed consent.

A screen failure is a consented subject who has been deemed ineligible on the basis of one or more eligibility criteria, or who has withdrawn consent prior to receiving any administration of study drug.

Potential subjects who meet all eligibility criteria will be scheduled for Baseline (Day 1) within the next 30 days. If a subject doesn't return for their Baseline visit within 30 days, all Screening procedures will need to be repeated prior to enrolling the subject.

The following procedures will be performed at the Screening visit:

- Obtain medical and ophthalmic history
- Obtain list of concomitant medications within at least the last 90 days (see Exclusion Criteria for medication exclusion periods)
- Collect vital signs
- Perform abbreviated physical exam
- Urine pregnancy test (for females of childbearing potential only)
- Obtain NL-BCVA using the ETDRS charts
- Perform slit-lamp biomicroscopy exam
- Measure IOP
- Perform dilated ophthalmoscopy exam

- Perform SD-OCT imaging
- Perform FAF imaging
- Obtain color fundus photographs
- Perform FA imaging
- Confirm initial eligibility

12.2. Baseline Visit (Day 1)

Eligible subjects will return for the Day 1 visit, at which time qualified subjects will be enrolled into the study.

If a subject in the Sentinel group is enrolled in the study and receives a dose level assignment, but is discontinued prior to receiving any administration of study drug, that subject may be replaced in order to ensure that the minimum number of subjects is enrolled per dose level.

The following procedures will be performed at the Day 1 visit:

- Update concomitant medications
- Collect vital signs
- Perform abbreviated physical exam
- Obtain NL-BCVA using the ETDRS charts
- For MTD subjects only: Obtain LL-BCVA using the ETDRS charts
- For MTD subjects only: Measure Contrast Sensitivity
- Perform slit-lamp biomicroscopy exam
- Measure pre-injection IOP
- Perform dilated ophthalmoscopy exam
- Perform SD-OCT imaging
- Perform FAF imaging
- Obtain color fundus photographs
- Confirm final eligibility
- Potential subjects who are eligible and willing to participate will be enrolled into the study, and assigned to the current dose level of study drug
- Subjects will receive their study drug implant in the study eye via IVT injection
- Perform a post-injection safety assessment of the study eye including IOP measurement
- Perform a post-injection study drug implant evaluation
- AE assessment
- Instruct subjects to contact the site study staff immediately if their treated eye becomes red, sensitive to light, painful, or develops a change in vision

12.3. Day 2 (24 Hours Post Injection ± 4 Hours)

The Day 2 (24 Hours Post Injection) visit may be conducted in the clinic or by phone, at the investigator's discretion. If a clinic visit is not performed, the subject will be contacted by phone to update concomitant medications and query for any potential AEs. If an AE is reported, the subject should return for an unscheduled visit per the judgement of the investigator.

If a clinic visit is conducted, the following procedures will be performed for the study eye only:

- Update concomitant medications
- Query for any AEs
- Obtain NL-BCVA using the ETDRS charts
- Perform slit-lamp biomicroscopy exam
- Measure IOP
- Perform dilated ophthalmoscopy exam
- Perform a study drug implant evaluation

If a phone visit is conducted, the following procedures will be performed:

- Update concomitant medications
- Query for any AEs

12.4. Week 1 (Day 8 ± 2 days)

The following procedures will be performed:

- Update concomitant medications
- Query for any AEs
- Collect vital signs
- Obtain NL-BCVA using the ETDRS charts
- Perform slit-lamp biomicroscopy exam
- Measure IOP
- Perform dilated ophthalmoscopy exam
- Perform a study drug implant evaluation

12.5. Week 4 (Day 29 ± 4 days)

The following procedures will be performed:

- Update concomitant medications
- Query for any AEs
- Collect vital signs
- Obtain NL-BCVA using the ETDRS charts
- Perform slit-lamp biomicroscopy exam
- Measure pre-injection IOP

- Perform dilated ophthalmoscopy exam
- Perform a study drug implant evaluation
- Perform SD-OCT imaging
- Perform FAF imaging
- Obtain color fundus photographs
- AE assessment

Note: For each of the 2 planned ascending dose levels of study drug in the Sentinel group, the DSC will review data once all enrolled subjects at that dose level have received their first implant and have completed the Week 4 visit. The DSC will review available safety data through at least Week 4, and if the DSC determines that the dose level is safe and tolerable, dosing will be allowed to escalate to the next dose level. Once dose-escalation has been completed or has been terminated due to the occurrence of dose-limiting toxicities (DLTs), the DSC will evaluate all available safety data through Week 4 of the highest dose administered. The DSC will then determine the MTD of OSA-1-100 for administration in the MTD group of up to 20 additional eyes.

12.6. Week 8, 12, 16, 20, 24, 28, 32 and 36 (± 4 days)

At all visits, the following procedures will be performed:

- Update concomitant medications
- Query for any AEs
- Collect vital signs
- Obtain NL-BCVA using the ETDRS charts
- Perform slit-lamp biomicroscopy exam
- Measure pre-injection IOP
- Perform dilated ophthalmoscopy exam
- Perform a study drug implant evaluation
- Perform SD-OCT imaging
- Perform FAF imaging
- Obtain color fundus photographs
- AE assessment

At Week 12, 24 and 36, the following additional procedures will be performed:

- For MTD subjects only: Obtain LL-BCVA using the ETDRS charts
- For MTD subjects only: Measure Contrast Sensitivity
- Subjects will receive a new study drug implant in the study eye via IVT injection
- Perform a post-injection safety assessment of the study eye including IOP measurement
- Perform a post-injection study drug implant evaluation

- Instruct subjects to contact the site study staff immediately if their treated eye becomes red, sensitive to light, painful, or develops a change in vision

12.7. Week 40, 44 and 48 Safety Follow-Up (± 4 days)

The following procedures will be performed:

- Update concomitant medications
- Query for any AEs
- Collect vital signs
- Obtain NL-BCVA using the ETDRS charts
- Perform slit-lamp biomicroscopy exam
- Measure IOP
- Perform dilated ophthalmoscopy exam
- Perform a study drug implant evaluation
- Perform SD-OCT imaging
- Perform FAF imaging
- Obtain color fundus photographs
- Perform FA imaging

At Week 48, the following additional procedures will be performed:

- For MTD subjects only: Obtain LL-BCVA using the ETDRS charts
- For MTD subjects only: Measure Contrast Sensitivity

12.8. Week 60 Safety Follow-Up (Day 421 ± 7 days)

The following procedures will be performed approximately 24 weeks after administration of the last scheduled study drug implant:

- Update concomitant medications
- Query for any AEs
- Collect vital signs
- Perform abbreviated physical exam
- Obtain NL-BCVA using the ETDRS charts
- For MTD subjects only: Obtain LL-BCVA using the ETDRS charts
- For MTD subjects only: Measure Contrast Sensitivity
- Perform slit-lamp biomicroscopy exam
- Measure IOP
- Perform dilated ophthalmoscopy exam
- Perform a study drug implant evaluation
- Perform SD-OCT imaging
- Perform FAF imaging

- Obtain color fundus photographs
- Perform FA imaging
- Study exit

Subjects with an ongoing SAE at this visit will be followed until the event is resolved or stabilized. The investigator will provide written reports of ongoing SAEs to the EC as required by the EC.

12.9. Study Exit

Subjects will be exited from the study at the conclusion of the Week 60 visit.

12.10. Early Termination Visit

Subjects who discontinue study drug or withdraw from the study early will undergo an Early Termination (ET) visit. At this visit, the following should be performed:

- Update concomitant medications
- Query for any AEs
- Collect vital signs
- Perform abbreviated physical exam
- Obtain NL-BCVA using the ETDRS charts
- For MTD subjects only: Obtain LL-BCVA using the ETDRS charts
- For MTD subjects only: Measure Contrast Sensitivity
- Perform slit-lamp biomicroscopy exam
- Measure IOP
- Perform dilated ophthalmoscopy exam
- Perform a study drug implant evaluation
- Perform SD-OCT imaging
- Perform FAF imaging
- Obtain color fundus photographs
- Perform FA imaging
- Study exit

Subjects who are exited from the study due to a SAE will be followed, if possible, until the event is resolved or stabilized. The investigator will provide written reports of ongoing SAEs to the EC as required by the EC.

12.11. Unscheduled Visits

Unscheduled visits may be necessary due to the occurrence of AEs or for other reasons. The investigator may examine a subject as often as is medically necessary while the subject is participating in the study, per the medical judgement of the investigator. Unscheduled follow-up assessments performed to monitor subject safety should be recorded as an Unscheduled visit.

At a minimum, NL-BCVA and the occurrence of any adverse events should be assessed for subjects reporting for an Unscheduled visit. Additional assessments may also be performed at the discretion of the investigator.

13. EARLY TERMINATION AND SUBJECT DISCONTINUATION

For each subject, the study will be considered completed when all assessments through Week 60 have been performed in accordance with the study protocol.

Any subject (or their legally authorized representative) may voluntarily discontinue the study at any time for any reason without prejudice. The investigator may elect to discontinue a subject for reasons unrelated to the study drug (e.g., failure to comply with the study protocol, missed visits, etc.) or for reasons related to the study drug (AE or unsatisfactory therapeutic response). In either event, reason(s) for discontinuation should be recorded on the eCRF.

Possible reasons for study discontinuation include the following:

- Significant protocol deviation (improper entry, prohibited therapy, subject noncompliance)
- AE necessitating discontinuation from the study
- The subject is lost to follow-up
- Subject decision unrelated to an AE (specify)
- Investigator administrative decision (specify)
- The study is terminated
- Other reason (specify)

If an enrolled subject chooses to discontinue his/her participation in the study, or the subject is discontinued by the investigator, the Study Exit eCRF shall be completed. Should a subject experience any difficulties requiring review between planned visits, or in case of early exit between planned visits, the Unscheduled Visit eCRF will be completed. Reason for such a visit will be recorded in the eCRF. Discontinued subjects will not be replaced. All subjects discontinued due to an AE will be followed until the event is resolved or considered medically stable by the investigator.

In case of a subject lost-to-follow-up, the investigator must do his best to contact the subject by any means available. If no response is obtained from the subject, the investigator is encouraged to contact one of the subject’s relatives or his/her family doctor. The evidence of these contacts must be recorded in the subject’s medical chart.

The Principal Investigator reserves the right to discontinue the study for any safety, ethical or administrative reason at any time.

14. STATISTICAL CONSIDERATIONS AND ANALYSIS

14.1. Sample size

The sample size was selected to place the minimum number of subjects at risk while providing sufficient safety information to move forward with subsequent larger studies.

14.2. Randomization

This is an open-label, ascending dose study. Randomization is not required.

14.3. General Considerations

All study parameters will be listed, and a selected list of parameters will be summarized descriptively for each dose level and overall. The sample size, mean, standard deviation, median, minimum and maximum will be presented for continuous analysis variables. Categorical variables will be presented as the count and percentage for each category. Statistical tests will be two-sided and evaluated at an α of 0.05, unless otherwise specified. No adjustment will be made for multiplicity.

Details of the statistical analyses to be performed for this study will be described in a separate Statistical Analysis Plan (SAP).

14.4. Missing Data

Missing data will not be imputed.

14.5. Study outcomes

Safety will be determined by assessing the incidence and severity of treatment emergent adverse events (TEAEs) related to the study drug and the occurrence of DLTs, as well as evaluating any changes in abbreviated physical exam findings, vital signs, visual acuity, IOP, slit lamp biomicroscopy findings and any findings from the dilated ophthalmoscopy examination.

Exploratory efficacy outcomes include:

- Change from baseline in drusen volume
- Change from baseline in peak/maximum drusen height
- Change from baseline in area of geographic atrophy
- Change from baseline in EZ thickness above peak drusen height
- Change from baseline in ONL thickness above peak drusen height
- Change from baseline in rate of photoreceptor (EZ) loss and EZ attenuation area in the study eye vs fellow eye
- Change from baseline in *en face* choroidal hypertransmission defects in study eye vs. fellow eye
- Change from baseline in BCVA
- Change from Baseline in low luminance BCVA (LL-BCVA) in MTD subjects
- Change from Baseline in Contrast Sensitivity in MTD subjects

14.6. Analysis Populations

The Safety Analysis Set will include all subjects who received a single injection of study drug. The Safety Analysis Set will be the primary population for evaluating all safety variables, exploratory efficacy variables and subject characteristics.

The DLT Evaluable Set will include all subjects who received a single administration of study drug and either completed the Week 4 visit or experienced a DLT during the 4-week post-injection observation period. The frequency and severity of DLTs will be analyzed based on the DLT Evaluable Set.

14.7. Statistical Hypotheses

For the exploratory endpoints, the change from baseline in the study eye will be compared to the change from baseline for the fellow eye. The null hypothesis is the difference between eyes, study eye minus fellow eye, is equal to 0. The alternative hypothesis is the difference in endpoint proportions is not equal to 0. The hypotheses can be expressed as:

H_0: D = 0

H_A: D ≠ 0

where D represents the difference in proportions, study eye minus fellow eye.

14.8. Statistical Analyses

Safety Analyses:
Safety will be determined by summarizing the incidence and severity of treatment emergent adverse events (TEAEs) related to the study drug. In addition, given that DLTs are used to determine the MTD of the study drug, the incidence of DLTs will be summarized.

Additional safety parameters to be evaluated include changes in abbreviated physical exam findings, vital signs, NL-BCVA, IOP, slit lamp biomicroscopy findings and any findings from the dilated ophthalmoscopy examination. Mean changes from baseline in vital signs, NL-BCVA and IOP will be summarized. Clinically meaningful changes from Baseline (Day 1) in NL-BCVA will also be summarized. A clinically meaningful change in NL-BCVA will be an increase or decrease of more than 10 letters (2 lines) on the ETDRS chart.

AEs will be mapped using MedDRA. All collected AE data will be listed by dose level and subject number. All AEs occurring after treatment with study drug on Day 1 (TEAEs) will be summarized by dose level and relationship to study drug. In addition, all serious adverse events (SAEs), including deaths (if any), will be listed separately and summarized. Finally, all ocular TEAEs will be listed and tabulated with events for the study eye and fellow eye summarized separately.

Efficacy Analyses:
Exploratory efficacy measures will be analyzed using paired statistics, paired t-test or Wilcoxon signed-rank test, comparing the study eye result to baseline or comparing the change from baseline between the study eye and the fellow eye.

14.9. FINAL REPORT

Once all study visits are complete and all data has been received and corrected, the study data will be locked. Once the data has been locked, a complete analysis of the results will be performed and a final report will be prepared.

15. ETHICAL AND REGULATORY CONSIDERATIONS

Investigator Responsibilities/Performance (Code of Conduct):

The investigator will ensure that the clinical study is conducted in accordance with the ethical principles that have their origin in the Declaration of Helsinki, GCP, ICH guidelines and any regional or national regulations as appropriate. The investigator will also ensure that subjects undergo no investigational procedures other than those described in this protocol and approved by the EC, except those procedures deemed necessary to protect the health and well-being of the subjects. Any such procedures will be documented in the subject's records and reported to the EC as required by the EC. If any additional requirements are imposed by the EC or regulatory authority, these requirements shall be followed.

Informed Consent Process and EC:

The investigator will obtain written informed consent from each subject prior to the initiation of any study-related activities. The obtaining of informed consent and the date of that informed consent is to be noted in the subject's medical notes. The site will submit the proposed informed consent to the EC. Copies of the informed consent form used in the study should contain the EC-approved stamp (if applicable) and version date. The following information should be provided in the informed consent:

- The study is research, and the purpose of the research;
- Purpose of the study, including the potential duration of participation by the subject and a description of the procedures;
- Possible side effects, risks or discomforts of the investigational drug;
- Reasonably expected benefits;
- The existence of alternative therapy options and/or alternatives to participating in the study;
- Confidentiality of records and possible inspection of records;
- Costs and compensation or treatment for injury;
- Person to contact regarding subject's rights;
- Person to contact for questions on research or research-associated injury;
- Participation is voluntary;
- The right to discontinue participation from the study at any time without disadvantage to the subject;
- Subject will be informed of new information learned during the study, which may affect the subject's decision to continue participation in the study.

The potential study subject or the subject's legally authorized representative will be allowed sufficient time to thoroughly read (or have read and explained to them), the informed consent form. The investigator will answer any questions that the potential study subject or their representative might have. If the potential study subject agrees to participate in the study, all required parties should sign and date the informed consent form. The original signed informed consent form will be retained by the site and a copy will be given to the potential study subject for their records. It should be noted in the potential study subject's medical record that the informed consent form was obtained prior to any study-related activities being conducted.

Investigator's Brochure:

Detailed documentation of data generated by the study drug will be made available to the investigator in the Investigator's Brochure and updated as appropriate during the course of the study. This information is confidential and remains the property of Osanni Bio.

16. INVESTIGATOR RESPONSIBILITIES

The investigator will ensure that the clinical study is conducted in accordance with Good Clinical Practice (GCP), International Conference on Harmonisation (ICH) guidelines and all regulatory and institutional requirements, including those for subject privacy, informed consent, Institutional Review Board/Ethics Committee (EC) approval and record retention.

The investigator is responsible for any expedited reports of SAEs or SUSARs to the EC as required by the EC.

17. CONCOMITANT MEDICATIONS AND THERAPIES

All concomitant medications including over-the-counter medications and nutritional supplements used within at least the 90 days (see Exclusion Criteria for medication exclusion periods) prior to the first visit will be documented. Any changes to concomitant medication use or initiation of new medications will be reported throughout the study. For the duration of the study, any therapy considered necessary for the maintenance of subject welfare may be given to the subject at the discretion of the investigator. Initiation of new medications or changes to concomitant medications should be recorded on the appropriate case report form.

17.1. PROHIBITED CONCOMITANT MEDICATIONS/PROCEDURES

The following medications and surgeries/procedures are prohibited as described below:

Medications

- Treatment with pegcetacoplan or avacincaptad pegol within 6 months prior to Screening or during the study in either eye.
- Previous intravitreal drug delivery (e.g., intravitreal corticosteroid injection, anti-angiogenic drugs or device implantation) in the study eye within 90 days prior to Screening.

- Participation in another investigational drug or device clinical trial within 90 days prior to Screening or during the study.

Surgeries/Procedures

- Any previous treatment with Visudyne® (verteporfin), external-beam radiation therapy (for intraocular conditions), or transpupillary thermotherapy in either eye.
- Any previous prophylactic subthreshold laser treatment for retinal disease in either eye.
- Any previous vitrectomy surgery, submacular surgery, or any vitreoretinal surgery in either eye within 12 months prior to Screening, or anticipated need for such surgery during the study.
- Ocular incisional surgery (including cataract surgery) in the study eye within 90 days prior to Screening, or anticipated need for such surgery during the study.

18. ADVERSE EVENT DEFINITIONS AND CHARACTERIZATION

18.1. Definitions

Adverse event (AE): An adverse event is defined as *any* untoward medical occurrence, unintended disease or injury or any untoward clinical signs (including an abnormal laboratory finding) whether related to the investigational drug or not. This includes events related to the investigational drug or any study procedures (any procedure in the study protocol).

Adverse Drug Reaction (ADR): An adverse event related to the use of an investigational drug. This includes any adverse event possibly, probably or definitely related to the investigational drug. It also includes any event that is a result of an investigational drug use error or intentional misuse.

Treatment-emergent AE (TEAE): Any AE with an onset from the time that the subject received study drug through 30 days after the last dose of study drug, whether or not it is considered causally related to the study drug.

18.2. Adverse Event Assessment and Documentation

All subjects who have been exposed to the study treatment will be evaluated for adverse events. All adverse events, regardless of severity and whether or not they are ascribed to the study treatment, will be recorded in the source documents and eCRFs using standard medical terminology.

All adverse events will be evaluated beginning with onset, and evaluation will continue until resolution is noted or until the investigator determines that the subject’s condition is stable. The investigator will take appropriate and necessary therapeutic measures required for resolution of the adverse event. Any medication necessary for the treatment of an adverse event must be recorded on the concomitant medication case report form.

Conditions or diseases that are present prior to initiation of study treatment and are chronic but stable should not be recorded on AE pages of the eCRF but should be recorded on the medical history. Changes in a chronic condition or disease that are consistent with natural disease progression do not need to be recorded as adverse events and should not be recorded on the AE pages of the eCRF.

Elevated IOP in the study eye following an intravitreal injection should be evaluated by the investigator as a potential AE if (1) at least 30 minutes after the injection, 2 or more follow-up IOP measurements (taken at least 15 minutes apart) are required to demonstrate stabilization to the pre-injection level (i.e., within 5 mm Hg of the pre-injection value); or (2) interventional therapy is required to return IOP to within 5 mm Hg of the pre-injection value.

Elective surgery or routine diagnostic procedures are not considered AEs.

Clinical study staff will start monitoring subjects for AEs from the time of written informed consent.

Whenever possible, recognized medical terms should be used when recording AEs. Colloquialisms and/or abbreviations should not be used. Only one medical concept, preferably a diagnosis instead of individual symptoms, should be recorded as the event (e.g. record congestive heart failure rather than dyspnea, rales and cyanosis). However, if a constellation of signs and/or symptoms cannot be medically characterized as a single diagnosis or syndrome at the time of reporting, each individual event should be recorded as a separate AE. A diagnosis that is subsequently established should be reported as follow-up information. However, signs and symptoms that are considered unrelated to an encountered syndrome or disease should be recorded as individual AEs (e.g. if congestive heart failure and severe headache are observed at the same time, each event should be recorded as an individual AE).

Adverse events occurring secondary to other events (e.g. sequelae) should be identified by the primary cause. A "primary" event, if clearly identifiable, should represent the most accurate clinical term to record as the AE event term. For example: Orthostatic hypotension ⇒fainting and fall to floor ⇒head trauma ⇒neck pain. The primary event is orthostatic hypotension and the sequelae are head trauma and neck pain.

18.3. Serious Adverse Events (SAE):

An AE should be classified as an SAE, and reported as such, if it meets one or more of the following criteria:

- It results in death (i.e., the AE actually causes or leads to death);
- It is life threatening (i.e., the AE places the subject at immediate risk of death at the time of the event; it does not refer to an event which hypothetically might have caused death if it were more severe);
- It requires or prolongs inpatient hospitalization. If a subject is hospitalized to undergo a medical or surgical procedure as a result of an AE, the event responsible for the procedure, not the procedure itself, should be recorded as the event. For example, if a subject is hospitalized to undergo coronary bypass surgery, the heart condition that necessitated the bypass should be recorded. Hospitalizations for diagnostic or elective

surgical procedures or hospitalizations required to allow outcome measurement for the study will not be recorded as SAEs;

- It results in persistent or significant disability/incapacity (i.e., the AE results in permanent or substantial disruption of the subject's ability to perform normal life functions);
- It is a congenital anomaly/birth defect in a neonate/infant born to a mother exposed to the investigational drug;
- It is an important medical event or reaction. Medical and scientific judgment should be exercised in deciding whether other situations should be considered serious such as important medical events that might not be immediately life-threatening or result in death or hospitalization but might jeopardize the subject or might require intervention to prevent one of the other outcomes listed in the definition above. Examples of such events are intensive treatment in an emergency room or at home for allergic bronchospasm, blood dyscrasias or convulsions that do not result in hospitalization, or development of drug dependency or drug abuse.

A planned hospitalization for a pre-existing condition without a serious deterioration in health, is not considered to be a serious adverse event. An emergency room visit lasting less than 24 hours is not considered hospitalization.

18.4. Serious and Unexpected Suspected Adverse Reaction

A serious and unexpected suspected adverse reaction (SUSAR) is any serious adverse event related or possibly related to a treatment if that effect, problem or death is not consistent with the nature, severity or degree of incidence described in the Investigator's Brochure or current product information. SUSARs also include any other unanticipated serious problem associated with a treatment that relates to the rights, safety or welfare of subjects.

18.5. Adverse Event Classification

Intensity/Severity: All adverse events should be graded on a 4-point scale (mild, moderate, marked, severe) for intensity/severity. These definitions are as follows:

- Mild: Transient discomfort; no medical intervention/therapy required and does not interfere with daily activities.
- Moderate: Low level of discomfort or concern with mild to moderate limitation in daily activities; some assistance may be needed; minimal or no medical intervention/therapy required.
- Marked: Considerable discomfort with limitation in daily activities; some assistance usually required; medical intervention/therapy usually required.
- Severe: Extreme discomfort and limitation in daily activities; significant assistance required; ***significant*** medical intervention/therapy required.

Note: There is a distinction between the severity and the seriousness of an adverse event. Severity is a measurement of intensity; thus, a severe reaction is not necessarily a serious adverse event (SAE). For example, a headache may be severe in intensity, but would not be serious unless it met one or more of the criteria for a serious adverse event.

Relationship to Study Drug: The study investigator will evaluate if the AE is related to the study drug. Relationship is defined in the following manner:

- Not related: Evidence indicates no plausible direct relationship to the study drug such that:
    - A clinically plausible temporal sequence is inconsistent with the onset of the event and study drug administration, and/or
    - A causal relationship of the study drug to the event is considered biologically implausible, and/or
    - The event is readily attributed to concurrent/underlying illness, or other drugs or procedures.
- Possibly related: Evidence indicates a possible relationship to the study drug such that:
    - The event occurs within a reasonable period of time relative to administration of the study drug that makes a causal relationship possible, and/or
    - There is a biologically plausible mechanism for the study drug causing or contributing to the event, and/or
    - The event could also be attributed to other causes such as other drugs, products, chemicals, concurrent/underlying disease, procedures, environment, etc.
- Related: Evidence indicates a reasonable temporal sequence of the event with the study drug administration, or the association of the event with study drug administration is unknown and the event is not reasonably explained by other conditions such that:
    - There is a clinically plausible time sequence between onset of the AE and study drug administration, and/or
    - The event is confirmed by stopping and/or restarting the study drug, and/or
    - There is a biologically plausible mechanism for the study drug causing or contributing to the event, and/or
    - The event cannot be reasonably attributed to concurrent/underlying disease, or other drugs or procedures.

AE Outcome: The clinical outcome of an AE will be characterized as follows:

- Resolved without sequelae
- Resolved with sequelae (specify)
- Ongoing (i.e. continuing at time of study exit or discontinuation)
- Death
- Unknown

Treatment Given for the AE: The clinical treatment or action taken from an AE will be characterized as follows:

- None
- Medical intervention
- Surgical intervention
- Other intervention

18.6. SAE AND SUSAR REPORTING

Serious adverse events (SAEs), whether or not considered drug related or expected, and serious unexpected suspected adverse reactions (SUSARs) are subject to expedited reporting requirements. The investigator is responsible for reporting SAEs and SUSARs to the EC as per their specified requirements. In addition, all SAEs, including SUSARs, must be reported to the manufacturer, Osanni Bio or its designee, as soon as possible and no later than 24 hours after the investigator first learns of the event.

For initial reports, investigators should record all case details that can be gathered within the reporting timeframe. The manufacturer contact information for reporting SAEs and SUSARs is below:

Eleonora Lad, MD PhD
Email: nora@osanni.bio and copy mark@osanni.bio

Phone: +1 650-796-8526
Alternate: Mark Holdbrook, +1 408-431-6613

Relevant follow-up information should be submitted to Osanni Bio or its designee as soon as it becomes available and/or upon request. For some events, Osanni Bio or its designee or the medical monitor may follow up with the site by telephone, fax, electronic mail and/or a monitoring visit to obtain additional case details deemed necessary to appropriately evaluate the event (e.g., hospital discharge summary, consultant report, or autopsy report). Reports relating to the subject’s subsequent medical course must be submitted to the EC until the event has subsided or, in case of permanent impairment, until the event stabilizes and the overall clinical outcome has been ascertained.

19. PREGNANCY

Women of childbearing potential (WOCBP) include any females who have experienced menarche and who have not undergone successful surgical sterilization (hysterectomy, bilateral tubal ligation or bilateral oophorectomy) or are not postmenopausal at least 2 years since last menses. WOCBP will be required to use designated methods of birth control during the course of the study. All women who are pregnant, nursing an infant, or planning a pregnancy during the duration of this study will be excluded from participation.

If a subject or investigator suspects that the subject may be pregnant prior to study drug administration, the study drug should be withheld until the results of pregnancy testing are available. If pregnancy is confirmed, the subject should not administer the study drug and should not be enrolled in the study.

If a female subject becomes pregnant during the study, the investigator will notify Osanni Bio (or designee) immediately after the pregnancy is confirmed. The investigator will (1) obtain a consent from the female subject for pregnancy follow-up and (2) follow the progress of the pregnancy to term. The investigator should document the outcome of the pregnancy and provide a copy of the documentation to Osanni Bio (or designee).

20. COMPLIANCE WITH PROTOCOL

Participation in the study will be summarized by presenting the number of subjects who received at least one dose of the study treatment, the number of subjects who completed the study per the protocol and the number of subjects who did not complete the study by reason of early termination/discontinuation.

21. PUBLICATION OF THE RESULTS AND PROTECTION OF COMMERCIAL SECRETS

The data generated in this clinical trial, all related information and any materials containing such data and information are the exclusive property of Osanni Bio and are confidential to Osanni Bio. The investigator or other study-related personnel may not disclose to anyone or use any data, information or materials related to this clinical trial without the express written consent of Osanni Bio.

Publication rights are governed by the Master Study Agreement between APEC and Osanni Bio.

22. DATA COLLECTION AND REPORTING

<u>Source Documentation:</u>

Source documentation for this study will be maintained to document the treatment and study course of each subject and to substantiate the integrity of the trial data submitted for review to the regulatory agencies. Source documentation will include, but may not be limited to, worksheets, hospital and/or clinic or office records documenting subject visits, including study and other treatments or procedures, medical/ophthalmologic history and physical examination information, laboratory and special assessment results, pharmacy records, drug accountability records and medical consultations.

<u>Case Report Forms:</u>

All historical data and required study observations obtained during the study will be entered in the subject's source documentation and transcribed to the appropriate eCRFs. eCRFs will be filled out completely for each subject in accordance with instructions from Osanni Bio or its designee. The investigator should ensure that all data on the eCRF is accurate and consistent with the source documents. Only the Principal Investigator or other designated sub-investigators may sign and date the designated eCRF page(s).

The eCRFs will be monitored and corrected on-site and if necessary, clarified on an ongoing basis. The site will be queried if any data, signatures or forms are missing or incorrect. Case report forms can be used as source documents where appropriate.

Monitoring and Inspection of Records:

The investigator must allow regular inspections of all study records including eCRFs, source documents and regulatory documents by the study monitor. This measure is to ensure that the study is carried out and documented in accordance with all regulatory requirements and the terms of this protocol.

In cooperation with the on-site staff, a designated monitor(s) will:

- Review all study documents to ensure compliance with the protocol;
- Review data recordings for each visit to verify the accuracy and completeness of the information on the eCRFs against appropriate source documents;
- Review study drug disposition and accountability records;
- Review adequacy of enrollment rate;
- Review required regulatory documentation.

Record Retention:

The investigator is responsible for maintaining adequate records to enable the conduct of the study to be fully documented. This includes:

- The Investigator's Brochure, signed protocol and amendments;
- Signed and dated informed consents per institutional policy;
- Signed, dated, and completed eCRFs and documentation of eCRF corrections;
- Notification of SAEs and related reports;
- All study drug dispensing and accountability logs;
- Shipping records of investigational drug and trial-related materials;
- Dated and documented EC protocol approvals and all correspondence between the investigator and EC;
- *Curriculum vitae* and current medical licenses for Principal Investigator and all sub-investigators;

Source documents:

The investigator is responsible for maintaining adequate case histories in the source documents of each subject. The following data will be noted by the investigator in the source medical records of each subject enrolled in the study:

- The subject is participating in this clinical trial;
- The date the informed consent form was signed;
- The dates of all study visits;
- Study drug kit numbers dispensed for the subject; and the

- Date of study completion or withdrawal.

The investigator must maintain a copy of all study documents for the period of time that is required by the regulatory authorities. No documents will be destroyed without prior written permission from Osanni Bio.

<u>Data Management:</u>

The data will be entered into a secure study database. The database will be electronically stored and a copy placed in back-up on a regular basis. All records are to be retained for at least 2 years after study completion, or as required by country regulations and the local EC. Any data inconsistencies will be resolved by a data clarification request and final resolution verified during site monitoring and study closure**.**

<u>Protocol Amendments:</u>

Any amendments to the protocol must be approved in advance by the investigator and the EC.

<u>Deviations from Clinical Investigation Plan:</u>

All deviations from the protocol should be clearly documented. Deviations conducted to protect a subject from possible hazard should be reported to the EC as soon as possible, as required by the EC.

23. REFERENCES

Abdelfattah, Nizar Saleh, Hongyang Zhang, David S. Boyer, and SriniVas R. Sadda. "Progression of Macular Atrophy in Patients with Neovascular Age-Related Macular Degeneration Undergoing Antivascular Endothelial Growth Factor Therapy." Retina (Philadelphia, Pa.) 36, no. 10 (October 2016): 1843–50.

Arditi, Aries. "Improving the Design of the Letter Contrast Sensitivity Test." Investigative Ophthalmology & Visual Science 46, no. 6 (June 1, 2005): 2225–29.

Bannerman, Augustine, Hanna Choi, Raviv Katz, Archana Nigalye, Grace Baldwin, Filippos Vingopoulos, Deeba Husain, Joan W Miller, Demetrios Vavvas, and John B Miller. "Drusen Volume Change in Patients with Age-Related Macular Degeneration Receiving High-Dose Atorvastatin." Investigative Ophthalmology & Visual Science 64, no. 8 (June 1, 2023): 295.

Barathi, Veluchamy, C. D. Luu, and I. P. Kaur. "Nano-Ocular Drops for Delivering a Statin to Posterior Eye Using SLN Delivery System: For Early and Late Form of Dry- AMD." Gold Cost, Queensland, Australia, 2023.

Bhisitkul, Robert B., Thais S. Mendes, Soraya Rofagha, Wayne Enanoria, David S. Boyer, SriniVas R. Sadda, and Kang Zhang. "Macular Atrophy Progression and 7-Year Vision Outcomes in Subjects from the ANCHOR, MARINA, and HORIZON Studies: The SEVEN-UP Study." American Journal of Ophthalmology 159, no. 5 (May 2015): 915-924.e2.

Boyer, David S., Ursula Schmidt-Erfurth, Menno van Lookeren Campagne, Erin C. Henry, and Christopher Brittain. “The Pathophysiology of Geographic Atrophy Secondary to Age-Related Macular Degeneration and the Complement Pathway as a Therapeutic Target.” Retina (Philadelphia, Pa.) 37, no. 5 (May 2017): 819–35.

Bressler, S. B., M. G. Maguire, N. M. Bressler, and S. L. Fine. “Relationship of Drusen and Abnormalities of the Retinal Pigment Epithelium to the Prognosis of Neovascular Macular Degeneration. The Macular Photocoagulation Study Group.” Archives of Ophthalmology (Chicago, Ill.: 1960) 108, no. 10 (October 1990): 1442–47.

Curcio, Christine A. “Soft Drusen in Age-Related Macular Degeneration: Biology and Targeting Via the Oil Spill Strategies.” Investigative Ophthalmology & Visual Science 59, no. 4 (March 20, 2018): AMD160–81.

Folgar, Francisco A., Eric L. Yuan, Monica B. Sevilla, Stephanie J. Chiu, Sina Farsiu, Emily Y. Chew, Cynthia A. Toth, and Age Related Eye Disease Study 2 Ancillary Spectral-Domain Optical Coherence Tomography Study Group. “Drusen Volume and Retinal Pigment Epithelium Abnormal Thinning Volume Predict 2-Year Progression of Age-Related Macular Degeneration.” Ophthalmology 123, no. 1 (January 2016): 39-50.

Gass, J. D. “Drusen and Disciform Macular Detachment and Degeneration.” Archives of Ophthalmology (Chicago, Ill.: 1960) 90, no. 3 (September 1973): 206–17.

Grunwald, Juan E., Maxwell Pistilli, Ebenezer Daniel, Gui-Shuang Ying, Wei Pan, Glenn J. Jaffe, Cynthia A. Toth, et al. “Incidence and Growth of Geographic Atrophy During 5 Years of Comparison of Age-Related Macular Degeneration Treatments Trials.” Ophthalmology 124, no. 1 (January 2017): 97–104.

Guymer, Robyn H., Paul N. Baird, Mary Varsamidis, Lucy Busija, Peter N. Dimitrov, Khin Zaw Aung, Galina A. Makeyeva, Andrea J. Richardson, Lyndell Lim, and Liubov D. Robman. “Proof of Concept, Randomized, Placebo-Controlled Study of the Effect of Simvastatin on the Course of Age-Related Macular Degeneration.” PloS One 8, no. 12 (2013): e83759.

Hadziahmetovic, Majda, and Goldis Malek. “Age-Related Macular Degeneration Revisited: From Pathology and Cellular Stress to Potential Therapies.” Frontiers in Cell and Developmental Biology 8 (January 25, 2021): 612812.

Holz, Frank G., Erich C. Strauss, Steffen Schmitz-Valckenberg, and Menno van Lookeren Campagne. “Geographic Atrophy: Clinical Features and Potential Therapeutic Approaches.” Ophthalmology 121, no. 5 (May 2014): 1079–91.

Johnson, Lincoln V., David L. Forest, Christopher D. Banna, Carolyn M. Radeke, Michelle A. Maloney, Jane Hu, Christine N. Spencer, et al. “Cell Culture Model That Mimics Drusen Formation and Triggers Complement Activation Associated with Age-Related Macular Degeneration.” Proceedings of the National Academy of Sciences of the United States of America 108, no. 45 (November 8, 2011): 18277–82.

Lindblad, Anne S., Patricia C. Lloyd, Traci E. Clemons, Gary R. Gensler, Frederick L. Ferris, Michael L. Klein, Jane R. Armstrong, and Age-Related Eye Disease Study Research Group. "Change in Area of Geographic Atrophy in the Age-Related Eye Disease Study: AREDS Report Number 26." Archives of Ophthalmology (Chicago, Ill.: 1960) 127, no. 9 (September 2009): 1168–74.

Linsenmeier, Robert A., and Hao F. Zhang. "Retinal Oxygen: From Animals to Humans." Progress in Retinal and Eye Research 58 (May 2017): 115–51.

Mitchell, Paul, Gerald Liew, Bamini Gopinath, and Tien Y. Wong. "Age-Related Macular Degeneration." Lancet (London, England) 392, no. 10153 (September 29, 2018): 1147–59.

Moore, David J., and Gillian M. Clover. "The Effect of Age on the Macromolecular Permeability of Human Bruch's Membrane." Investigative Ophthalmology & Visual Science 42, no. 12 (November 1, 2001): 2970–75.

Ortman, Jennifer M., Victoria A. Velkoff, and Howard Hogan. "An Aging Nation: The Older Population in the United States." Current Population Reports. U.S. Department of Commerce: Economics and Statistics Administration: U.S. Census Bureau, May 2014.

Osanni Bio Internal Study. Study 23-1475. "A 24-Week Intravitreal Ocular Distribution Study of OSA-1-100r in Dutch Belted Rabbits."

Rein, David B., John S. Wittenborn, Zeb Burke-Conte, Rohit Gulia, Toshana Robalik, Joshua R. Ehrlich, Elizabeth A. Lundeen, and Abraham D. Flaxman. "Prevalence of Age-Related Macular Degeneration in the US in 2019." JAMA Ophthalmology 140, no. 12 (December 1, 2022): 1202–8.

Sadigh, Sam, Artur V. Cideciyan, Alexander Sumaroka, Wei Chieh Huang, Xunda Luo, Malgorzata Swider, Janet D. Steinberg, Dwight Stambolian, and Samuel G. Jacobson. "Abnormal Thickening as Well as Thinning of the Photoreceptor Layer in Intermediate Age-Related Macular Degeneration." Investigative Ophthalmology & Visual Science 54, no. 3 (March 5, 2013): 1603–12.

Sivaprasad, Sobha, Elizabeth Tschosik, Robyn Guymer, Audrey Kapre, Ivan Suner, Antonia Joussen, Paolo Lanzetta, and Daniela Ferrara. "Living with Geographic Atrophy: An Ethnographic Study." Ophthalmology and Therapy 8 (January 31, 2019).

Stancu, C., and A. Sima. "Statins: Mechanism of Action and Effects." Journal of Cellular and Molecular Medicine 5, no. 4 (2001): 378–87.

VanNewkirk, M. R., M. B. Nanjan, J. J. Wang, P. Mitchell, H. R. Taylor, and C. A. McCarty. "The Prevalence of Age-Related Maculopathy: The Visual Impairment Project." Ophthalmology 107, no. 8 (August 2000): 1593–1600.

24. APPENDIX A: EXAMS AND PROCEDURES

24.1. Informed Consent

Prior to performing any study specific procedures, the potential study subject (or the subject's legally authorized representative) will be provided with the current EC approved informed consent form (ICF) and be given adequate time to review and ask any questions about the study and/or the ICF before written informed consent is obtained. The consenting process must be conducted in accordance with GCP, ICH guidelines and local regulatory and/or EC requirements.

The potential study subject (or the subject's legally authorized representative) will sign and date the informed consent form in the presence of a witness, if required. The Principal Investigator or his designee will also sign and date the consent form. The original signed consent will be retained with the potential study subject's medical records, and a copy will be provided to the potential study subject.

If the protocol is amended or new safety information becomes available, and revisions are required to the ICF, the subject (or subject's legally authorized representative) will be required to sign the updated EC approved ICF.

24.2. Vital Signs

Vital signs should be taken after the subject has been resting in a seated position for at least 5 minutes. Vital signs measurements will include blood pressure (systolic and diastolic in millimeter of mercury [mmHg]), heart rate (beats/minute), respiratory rate (breaths/minute), and temperature (°C).

24.3. Abbreviated Physical Exam

Abbreviated physical examination will include assessments of general appearance, cardiovascular, respiratory, neck/HEENT (head, eyes, ears, nose and throat) and dermatologic. Additional assessments should be completed per the judgment of the investigator. All clinically significant (CS) abnormalities will be recorded. A change from normal (within normal limits) to Abnormal-CS for any physical exam category will be considered clinically significant.

Height and weight will be measured at the Screening visit only.

24.4. Best-Corrected Visual Acuity (BCVA)

Normal luminance BCVA (NL-BCVA) using manifest refraction will be measured for each eye, prior to dilating eyes, using standard Early Treatment Diabetic Retinopathy Study (ETDRS) retro-illuminated charts in normal luminance lighting.

Low luminance BCVA (LL-BCVA) using manifest refraction will be measured after NL-BCVA is measured in normal luminance. A 2.0 log unit neutral density filter (i.e., a filter that lowers luminance by 100 times) will be placed over the best correction for each eye in order to perform LL-BCVA using the same standard Early Treatment Diabetic Retinopathy Study (ETDRS) retro-illuminated charts.

Both NL-BCVA and LL-BCVA will be recorded as the total letter score for each eye.

Visual acuity testing will be performed by designated site staff and should occur prior to any examination requiring contact with the eye. In order to provide standardization and well-controlled measurements of NL-BCVA and LL-BCVA during the study, all measurements for an individual study subject must be done consistently throughout the study using the same lighting conditions, correction, chart type, and measurement procedure.

Note: A clinically significant decrease from Baseline (Day 1) in NL-BCVA (defined as a decrease of 11 or more ETDRS letters) should be evaluated by the investigator as a potential AE.

Refraction and BCVA testing procedures will be described in a separate Visual Function Testing Procedures Manual.

24.5. Contrast Sensitivity

Contrast sensitivity will be measured for each eye, prior to dilating eyes, using Mars contrast sensitivity charts (Arditi, 2005). The contrast sensitivity score for each eye will be recorded.

Contrast sensitivity testing will be performed by designated site personnel and should occur after BCVA testing but before any examination requiring contact with the eye. In order to provide standardization and well-controlled measurements of contrast sensitivity during the study, all measurements at a single site must be consistently done using uniform illumination, correction, chart type, and measurement procedure for an individual subject throughout the study.

Instructions for measuring contrast sensitivity will be provided in a separate Visual Function Testing Procedures Manual.

24.6. Slit Lamp Biomicroscopy

A slit lamp examination of the anterior segment of the eye will be performed prior to pupil dilation, and the following structures will be assessed for pathology:

- Eyelids (i.e., Meibomian glands, lid margin, puncta, lashes)
- Cornea
- Conjunctiva
- Anterior chamber
- Iris
- Lens

For the biomicroscopy variables, all clinically significant (CS) and non-clinically significant (NCS) abnormalities will be recorded. A change from normal (within normal limits) to Abnormal-CS or from Abnormal-NCS to Abnormal-CS for any biomicroscopy finding will be considered clinically significant.

24.7. Intraocular Pressure (IOP)

Intraocular pressure will be measured using a contact applanation tonometer, such as the Goldmann or Perkins, consistent with the site's standard clinical practice for measuring IOP. The same tonometry method should be used throughout an individual subject's participation in the study. The subject should be sitting comfortably at the slit lamp, at the appropriate height, with their chin on the rest and their forehead against the headband (or in a chair with their head supported, if using the Perkins tonometer). IOP measurements will be recorded in mmHg (e.g., 19 mmHg).

At study drug injection visits, IOP will be measured by designated site staff in both eyes before the injection, and then in the study eye only at least 30 minutes after the injection. Subsequent post-injection IOP measurements in the study eye may be performed as needed, as part of the post-injection safety assessment.

The tonometer must be calibrated for accuracy before the first subject in the study at the site undergoes the first examination, and at least once every month thereafter, until the last subject at the site has exited the study. Tonometer calibration may be completed by any qualified site staff and should be performed in accordance with the manufacturer's specifications. Only an adequately calibrated instrument should be used for IOP measurements.

24.8. Dilated Ophthalmoscopy Examination

An examination of the posterior segment of the eye will be performed using indirect ophthalmoscopy. Pupils should be dilated adequately to perform a thorough fundus evaluation. The following structures will be assessed for pathology:

- Vitreous
- Optic nerve
- Macula
- Retina
- Choroid

All clinically significant (CS) and non-clinically significant (NCS) abnormalities will be recorded. A change from normal (within normal limits) to Abnormal-CS or from Abnormal-NCS to Abnormal-CS for any ophthalmoscopy finding will be considered clinically significant.

With pupillary dilation, the subject will be informed about possible light sensitivity lasting for hours and will be instructed to wear dark glasses in bright light conditions to reduce sensitivity.

24.9. Study Drug Implant Evaluation

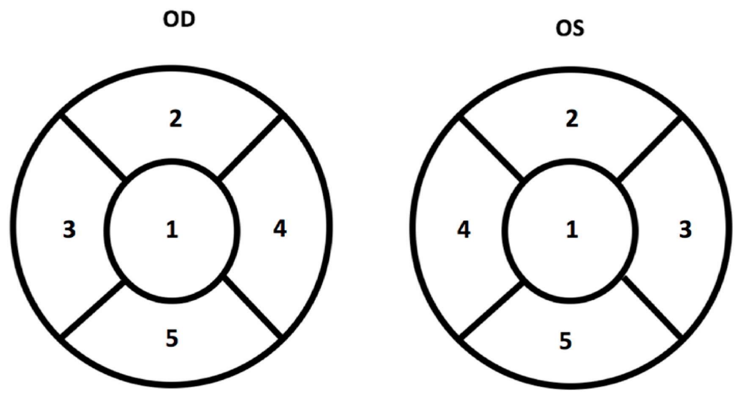
Following the dilated ophthalmoscopy exam, an evaluation of the OSA-1-100 implant will be performed. This will include the following implant assessments:

- Presence (implant observed or implant not observed)
- General location (anterior, posterior or vitreous chamber)
- Ocular region (refer to the Ocular Regions diagram below)

- Visual appearance (e.g., implant is intact, remnants of implant observed)

Ocular Regions:

1. Central/Macular region
2. Superior quadrant
3. Temporal quadrant
4. Nasal quadrant
5. Inferior quadrant



24.10. SPECTRAL-DOMAIN OPTICAL COHERENCE TOMOGRAPHY (SD-OCT)

SD-OCT will be performed for each eye per the Schedule of Visits and the Reading Center manual of procedures. The study site staff designated to perform SD-OCT imaging must be certified by the Reading Center prior to obtaining images of any study subjects.

24.11. COLOR FUNDUS PHOTOGRAPHY (CFP)

CFP will be performed for each eye per the Schedule of Visits and the Reading Center manual of procedures. The study site staff designated to perform CFP imaging must be certified by the Reading Center prior to obtaining images of any study subjects.

24.12. FUNDUS AUTOFLUORESCENCE (FAF)

FAF will be performed for each eye per the Schedule of Visits and the Reading Center manual of procedures. When FAF and FA are performed at the same visit, FAF must be performed before FA. The study site staff designated to perform FAF imaging must be certified by the Reading Center prior to obtaining images of any study subjects.

24.13. FLUORESCEIN ANGIOGRAPHY (FA)

FA will be performed for each eye per the Schedule of Visits and the Reading Center manual of procedures. FA must be performed after all other scheduled imaging has been completed. The study site staff designated to perform FA imaging must be certified by the Reading Center prior to obtaining images of any study subjects.